A Proof of Quadratic Reciprocity from a Symplectic Similitude

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Abstract

We give a self-contained proof of quadratic reciprocity using the augmentation module of a cyclic group of odd prime order p over $\mathbf{F}_q$, where $q \neq p$ is an odd prime. The cyclic shift T defines an invertible skew-adjoint operator $C = T - T^{-1}$, and hence a nondegenerate alternating form. A multiplier permutation P satisfies $PCP^{-1} = C^q$. The correcting operator $C^{(q-1)/2}P$ is consequently a symplectic similitude with an explicitly determined multiplier. Comparing its determinant with the sign of the multiplier permutation yields the reciprocity law. A division-free Pfaffian argument proves the determinant identity for symplectic similitudes in every odd characteristic.

1 Introduction

For an odd prime ℓ and an integer a not divisible by ℓ, the Legendre symbol $\left(\frac{a}{\ell}\right)$ is 1 if the image of a in $\mathbf{F}_\ell^\times$ is a square, and is -1 otherwise. Quadratic reciprocity compares the two residue questions obtained by interchanging a pair of distinct odd primes. For Gauss's original treatment, see [2, Sectio IV].

Theorem 1.1 (Quadratic reciprocity)**.** *For distinct odd primes p and q,*

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{(p-1)(q-1)/4}. \tag{1.1}$$

The purpose of this paper is to explain how the sign in (1.1) arises from an alternating form. We work over $\mathbf{F}_q$ on the $(p-1)$-dimensional augmentation module

$$V = \left\{ (x_i)_{i \in \mathbf{Z}/p\mathbf{Z}} \in \mathbf{F}_q^p : \sum_{i \in \mathbf{Z}/p\mathbf{Z}} x_i = 0 \right\}.$$

The dot product gives a nondegenerate symmetric form B on V. If T denotes the cyclic shift, then

$$C = T - T^{-1}, \qquad \omega(x, y) = B(x, Cy)$$

give an invertible operator and a nondegenerate alternating form. The permutation $Pe_i = e_{qi}$ preserves B, but its interaction with ω is controlled by the relation $PCP^{-1} = C^q$. Multiplying P by $C^{(q-1)/2}$ compensates for this change of form.

More precisely, put $m = (p-1)/2$ and $r = (q-1)/2$. We will show that

$$M = C^r P, \qquad \omega(Mx, My) = (-1)^r \omega(x, y).$$

The determinant of a symplectic similitude with multiplier μ in dimension $2m$ is μ^m. On the other hand, $\det C = p$, while the sign interpretation of the Legendre symbol gives $\det P = \left(\frac{q}{p}\right)$. These two ways of computing $\det M$ lead directly to (1.1).

All ingredients needed for the proof are established below. We assume familiarity with finite-dimensional linear algebra, including characteristic polynomials, determinants, and bilinear forms.

2 Quadratic characters and permutation determinants

We first recall Euler's criterion and the permutation-sign lemma. Their proofs will also clarify the distinction between an integer sign and its image in a finite field.

Lemma 2.1 (Euler's criterion)**.** *If ℓ is an odd prime and $a \in \mathbf{F}_\ell^\times$, then*

$$a^{(\ell-1)/2} = \left(\frac{a}{\ell}\right) \quad \text{in } \mathbf{F}_\ell,$$

where the symbol on the right is identified with its image in the field.

Proof. Multiplication by a permutes the nonzero elements of $\mathbf{F}_\ell$. Taking their product and cancelling gives $a^{\ell-1} = 1$.

The squaring map on $\mathbf{F}_\ell^\times$ has fibers of size two, since $x^2 = y^2$ implies $x = y$ or $x = -y$. There are therefore exactly $(\ell-1)/2$ nonzero squares. Each is a root of $X^{(\ell-1)/2} - 1$, and the degree bound for the number of roots shows that these are all its roots. Finally, $a^{\ell-1} = 1$ implies $a^{(\ell-1)/2} \in \{1, -1\}$. It is 1 precisely for squares, proving the assertion. □

Lemma 2.2 (Zolotarev's sign lemma)**.** *Let ℓ be an odd prime and let $\ell \nmid a$. The permutation*

$$\sigma_a : \mathbf{Z}/\ell\mathbf{Z} \longrightarrow \mathbf{Z}/\ell\mathbf{Z}, \qquad i \longmapsto ai$$

has integer sign

$$\operatorname{sgn}(\sigma_a) = \left(\frac{a}{\ell}\right).$$

Proof. For $z_0, \ldots, z_{\ell-1} \in \mathbf{F}_\ell$, write

$$\Delta(z_0, \ldots, z_{\ell-1}) = \prod_{0 \le i < j \le \ell-1} (z_j - z_i).$$

This is the Vandermonde determinant, and is alternating in its arguments. In particular, $\Delta(0, 1, \ldots, \ell-1) \neq 0$. Permuting these arguments by σ_a, and also evaluating by scaling them by a, gives

$$\operatorname{sgn}(\sigma_a)\Delta(0, 1, \ldots, \ell-1) = a^{\ell(\ell-1)/2}\Delta(0, 1, \ldots, \ell-1).$$

After cancellation, Lemma 2.1 gives

$$\operatorname{sgn}(\sigma_a) = \left(a^{(\ell-1)/2}\right)^\ell = \left(\frac{a}{\ell}\right) \quad \text{in } \mathbf{F}_\ell.$$

Both sides are integer signs. Since ℓ is odd, their images in $\mathbf{F}_\ell$ are distinct unless the signs are equal. □

The interpretation in Lemma 2.2 is classical; see Brunyate and Clark [1] for its history and extensions. We will apply it with $\ell = p$ and $a = q$. The sign is computed over $\mathbf{F}_p$, but the resulting integer 1 or -1 can then be used as the determinant of the same permutation matrix over $\mathbf{F}_q$.

3 Alternating forms and a correcting power

Let U be a vector space of dimension $2m$ over a field k of odd characteristic. A bilinear form ω is alternating if $\omega(x,x)=0$ for all x, and is nondegenerate if $\omega(x,y)=0$ for every y forces $x=0$. An invertible operator S is a *symplectic similitude* with multiplier $\mu \in k^\times$ if

$$\omega(Sx,Sy)=\mu\,\omega(x,y).$$

Lemma 3.1 (Determinant of a symplectic similitude)**.** *If ω is nondegenerate and S is a symplectic similitude with multiplier μ, then*

$$\det S=\mu^m.$$

Proof. We construct a nonzero alternating $2m$-linear form without dividing by any integer. Let $\mathcal{P}_m$ be the set of partitions of $\{1,\dots,2m\}$ into unordered pairs. Write a partition uniquely as

$$\{\{i_1,j_1\},\dots,\{i_m,j_m\}\},\qquad i_s<j_s,\quad i_1<\cdots<i_m,$$

and let $\varepsilon(\mathcal{P})$ be the sign of the permutation $(i_1,j_1,\dots,i_m,j_m)$. Define

$$\Omega(v_1,\dots,v_{2m})=\sum_{\mathcal{P}\in\mathcal{P}_m}\varepsilon(\mathcal{P})\prod_{s=1}^{m}\omega(v_{i_s},v_{j_s}). \tag{3.1}$$

Each vector occurs once in each summand, so Ω is multilinear. Reindexing the pairings shows that permuting its arguments changes the value by the sign of that permutation: reversing a pair contributes the sign from the skew-symmetry of ω, while interchanging two whole pairs is an even permutation. In odd characteristic this also implies that Ω vanishes when two arguments coincide. Thus Ω is alternating.

For completeness, a symplectic basis exists by an elementary induction. Choose $e\neq 0$, and then choose f with $\omega(e,f)=1$. The restriction to $H=\operatorname{span}\{e,f\}$ is nondegenerate, and $U=H\oplus H^\perp$, with nondegenerate restriction to $H^\perp$. Repeating gives an ordered basis $(e_1,f_1,\dots,e_m,f_m)$ in which the only nonzero pairings of distinct basis vectors are $\omega(e_i,f_i)=1$ and their negatives. Only one summand in (3.1) survives on this basis, so

$$\Omega(e_1,f_1,\dots,e_m,f_m)=1.$$

In particular, $\Omega\neq 0$.

Every alternating form of top degree transforms by the determinant; this follows directly by expanding each argument in a basis. Consequently,

$$\Omega(Sv_1,\dots,Sv_{2m})=(\det S)\Omega(v_1,\dots,v_{2m}).$$

The similitude relation, applied to each of the m factors in every summand of (3.1), also gives

$$\Omega(Sv_1,\dots,Sv_{2m})=\mu^m\Omega(v_1,\dots,v_{2m}).$$

Evaluating on the symplectic basis proves the result. □

Now let B be a nondegenerate symmetric bilinear form on U. For an operator A, its B-adjoint $A^\dagger$ is defined by

$$B(Ax,y)=B(x,A^\dagger y).$$

Nondegeneracy makes the adjoint unique. For operators A,D, it satisfies $(AD)^\dagger=D^\dagger A^\dagger$. An operator C is skew-adjoint if $C^\dagger=-C$, and an invertible operator P is a B-isometry if $P^\dagger=P^{-1}$.

Proposition 3.2. *Let $k = \mathbf{F}_Q$ be a finite field of odd order, and let (U, B) be a nondegenerate symmetric bilinear space of dimension $2m$. Suppose that $C \in \mathrm{GL}(U)$ is B-skew-adjoint and that P is a B-isometry satisfying*

$$PCP^{-1} = C^Q.$$

Put $s = (Q-1)/2$. Then

$$\omega(x, y) = B(x, Cy)$$

is a nondegenerate alternating form, and $M = C^s P$ is a symplectic similitude with multiplier $(-1)^s$. In particular,

$$(\det C)^s \det P = (-1)^{ms}. \tag{3.2}$$

Proof. Symmetry of B and skew-adjointness of C give

$$\omega(y, x) = B(y, Cx) = B(Cx, y) = B(x, -Cy) = -\omega(x, y).$$

Since the characteristic is odd, ω is alternating. Invertibility of C and nondegeneracy of B imply nondegeneracy of ω.

For the claimed multiplier, compute

$$\begin{aligned} M^\dagger C M &= P^{-1}(C^s)^\dagger C C^s P \\ &= (-1)^s P^{-1} C^{2s+1} P \\ &= (-1)^s P^{-1} C^Q P \\ &= (-1)^s C. \end{aligned}$$

Hence

$$\omega(Mx, My) = B(x, M^\dagger C M y) = (-1)^s \omega(x, y).$$

Lemma 3.1 now gives $\det M = (-1)^{ms}$, while multiplicativity of determinants gives $\det M = (\det C)^s \det P$. □

The calculation explains the role of the exponent $(Q-1)/2$. Passing C^s through the symmetric form produces the sign $(-1)^s$ and raises the middle operator C to the power $2s+1 = Q$. The conjugacy relation with P then returns this operator to C.

4 The cyclic augmentation module

Fix distinct odd primes p, q, and set

$$k = \mathbf{F}_q, \qquad m = \frac{p-1}{2}, \qquad r = \frac{q-1}{2}.$$

Let W have basis $\{e_i : i \in \mathbf{Z}/p\mathbf{Z}\}$, and give it the dot product

$$B(x, y) = \sum_{i \in \mathbf{Z}/p\mathbf{Z}} x_i y_i.$$

Write $\mathbf{1} = \sum_i e_i$, $L = k\mathbf{1}$, and

$$V = \left\{ \sum_i x_i e_i : \sum_i x_i = 0 \right\}.$$

Define permutation operators on W by

$$Te_i = e_{i+1}, \qquad Pe_i = e_{qi}.$$

Indices are taken modulo p. Both T and P fix $\mathbf{1}$, preserve V, and preserve B. We use the same letters for their restrictions to V.

Proposition 4.1. *The restriction of B to V is nondegenerate. On V, the operator $C = T - T^{-1}$ is skew-adjoint and satisfies*

$$\det C = p, \qquad PCP^{-1} = C^q, \qquad \det P = \left(\frac{q}{p}\right), \tag{4.1}$$

where integer scalars and signs are reduced into k. In particular, C is invertible.

Proof. Since $B(x, \mathbf{1}) = \sum_i x_i$, we have $V = L^\perp$. Moreover, $B(\mathbf{1}, \mathbf{1}) = p \neq 0$ in k. Thus

$$W = L \perp V,$$

and the restriction of B to V is nondegenerate. Its dimension is $p-1 = 2m$. Since T is an isometry,

$$C^\dagger = (T - T^{-1})^\dagger = T^{-1} - T = -C.$$

To compute the determinant, note first that T is a p-cycle on the basis of W, so $\det(T|_W) = (-1)^{p-1} = 1$. As $T|_L = \mathrm{id}$, this gives $\det(T|_V) = 1$. The operator T^2 is also a p-cycle, because p is odd. Its characteristic polynomial on W is $X^p - 1$: in a cyclically ordered basis its matrix is the companion matrix of this polynomial. Splitting off L gives

$$\det(X\,\mathrm{id} - T^2|_V) = \frac{X^p - 1}{X - 1} = 1 + X + \cdots + X^{p-1}.$$

Evaluating at $X = 1$, we obtain $\det(\mathrm{id} - T^2|_V) = p$. Since $\dim V$ is even and $C = T^{-1}(T^2 - \mathrm{id})$, it follows that

$$\det C = (\det T)^{-1} \det(T^2 - \mathrm{id}) = p.$$

This is nonzero in k, proving invertibility.

A calculation on basis vectors gives $PTP^{-1} = T^q$. Because T and T^{-1} commute and k has characteristic q, the binomial identity yields

$$PCP^{-1} = T^q - T^{-q} = (T - T^{-1})^q = C^q.$$

The exponent here denotes an ordinary power of an operator.

Finally, on W the determinant of P is the sign of the permutation $i \mapsto qi$ of $\mathbf{Z}/p\mathbf{Z}$. By Lemma 2.2, this integer sign is $\left(\frac{q}{p}\right)$. Since $P|_L = \mathrm{id}$, the determinant on V is the same. □

5 Proof of quadratic reciprocity

We can now apply the general construction. The symmetric space in Proposition 4.1 and its operators C, P satisfy all hypotheses of Proposition 3.2, with $Q = q$.

Proof of Theorem 1.1. On V, define

$$\omega(x, y) = B(x, Cy), \qquad M = C^r P.$$

Proposition 3.2 gives

$$\omega(Mx, My) = (-1)^r \omega(x, y), \qquad \det M = (-1)^{mr}.$$

The determinants in (4.1) give a second expression,

$$\det M = (\det C)^r \det P = p^r \left(\frac{q}{p}\right).$$

By Euler's criterion in $\mathbf{F}_q$, $p^r = \left(\frac{p}{q}\right)$. Therefore

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{mr} \quad \text{in } \mathbf{F}_q.$$

Both sides are signs, and q is odd, so equality in $\mathbf{F}_q$ implies equality as integers. Substituting $mr = (p-1)(q-1)/4$ proves (1.1). □

The reciprocity sign is thus the top-degree effect of the multiplier $(-1)^r$ on a symplectic space of dimension $2m$. This is the point at which the two primes interact in the determinant calculation.

References

[1] A. Brunyate and P. L. Clark, *Extending the Zolotarev–Frobenius approach to quadratic reciprocity*, The Ramanujan Journal **37** (2015), 25–50. doi:10.1007/s11139-014-9635-y.

[2] C. F. Gauss, *Disquisitiones arithmeticae*, Gerh. Fleischer, Jun., Leipzig, 1801, Sectio IV. Digitized original, ETH-Bibliothek Zürich.

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